

Stochastic Modeling and Optimal Control for Algorithmic Trading

From market microstructure and statistical inference to adaptive trading policies

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Preface

Algorithmic trading is often taught as a collection of models: one model for impact, another for volume, a dynamic program for liquidation, a classifier for fills, and perhaps a reinforcement-learning agent at the end. This book takes a different view. These are components of one feedback system.

The trader observes an incomplete and noisy state of the market, estimates how prices and liquidity may evolve, chooses actions that change both inventory and future trading conditions, and learns from fills and market response. Statistics describes what can be inferred; stochastic modeling describes how uncertainty evolves; control theory describes how beliefs should become decisions.

Intuition

The unifying question is not “Which algorithm should I use?” It is: **What is the state, what can I observe, what can I influence, what does my action reveal or disturb, and which cost matters for the decision?**

The opening five chapters form a self-contained execution-and-impact track. They emphasize the questions that arise in execution research and market-impact interviews: why costs increase with urgency, what an impact coefficient actually identifies, why realized slippage is endogenous, when a smooth schedule is optimal, and how alpha decay, liquidity forecasts, and constraints change the answer.

The intended rhythm is intuition $\rightarrow$ assumptions $\rightarrow$ derivation $\rightarrow$ empirical test $\rightarrow$ control implication. “Laws” of market impact are treated as conditional empirical regularities rather than universal constants. Machine-learning models are judged not only by predictive accuracy but by calibration, stability, and the decisions they induce.

0.1 How to use the book

- For an execution interview, read Chapters 1–5 in order and solve every interview check without looking back.
- For research practice, reproduce the empirical diagnostics and ask what selection mechanism generated the sample.
- For theory, re-derive results after changing one assumption at a time.
- For production modeling, trace the units and information set of every feature before considering model class.

This is a working manuscript. The first priority is conceptual and mathematical depth in execution and market impact; later parts expand the same framework to order placement, routing, market making, portfolio execution, and learning.

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2. Market Microstructure as the Controlled System
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4. Market-Impact Models and Estimation
5. Optimal Liquidation and Participation Control

1 Trading as Inference and Control

A parent order is not merely a quantity to split. It is a sequence of decisions made with partial information while the system changes. Prices move, liquidity appears and disappears, a signal decays, fills reveal information, and the trader's own actions may change the market. This is the natural language of stochastic control.

This chapter develops a reusable representation before choosing a particular price model or optimizer. The representation is more important than any one closed-form solution: it tells us which forecasts are relevant, which costs are missing, and when a backtest is answering the wrong question.

1.1 The canonical execution problem

Suppose a buy order of $Q > 0$ shares arrives at time 0 and must be completed by T . Let X_t be remaining inventory, so

$$X_0 = Q, \quad X_T = 0, \quad dX_t = -v_t dt,$$

where the trading rate $v_t \geq 0$ has units shares per unit time. We use signed quantities when both sides are needed, but fixing the side initially prevents sign conventions from hiding the economics.

At time t , the algorithm does not observe “the market” as a scalar. A useful state vector might be

$$Z_t = (X_t, t, m_t, s_t, D_t, V_t, \sigma_t, a_t, q_t, r_t).$$

Here m_t is midprice; s_t is spread; D_t summarizes depth; V_t is forecast market volume; σ_t is volatility; a_t is an alpha or drift state; q_t is queue position; and r_t is a market-resilience or transient-impact state. Not every model needs every component. Choosing the smallest state that preserves the decision-relevant future is a central modeling task.

Definition: admissible policy

A policy $\pi = \{\pi_t\}$ maps the information available at time t into an action: $u_t = \pi_t(\mathcal{F}_t)$. It is admissible if it respects information timing, trading constraints, and integrability conditions. A backtest that uses a feature unavailable at decision time evaluates a non-admissible policy.

The action u_t can be a market-order rate, a vector of venue allocations, limit prices and sizes, cancellation decisions, or a mixture. The disturbance includes future price innovations, other agents' order flow, fills, volume, and latency.

1.2 State, observation, and belief

The true economic state is rarely observed. Displayed depth omits hidden and latent liquidity. The origin and remaining size of a metaorder are unobserved. The current volatility regime is estimated with error. It is therefore useful to separate

$$Z_{t+1} = F(Z_t, u_t, \varepsilon_{t+1}), \quad Y_t = H(Z_t, \eta_t),$$

where Z_t is latent state, Y_t is observed data, and ε, η are state and observation noise. If the observation is not sufficient for control, the state becomes a **belief** $b_{t(z)} = P(Z_t = z \mid \mathcal{F}_t)$. This converts a partially observed problem into a fully observed problem on distributions.

Intuition

“State” does not mean every column in the feature store. It means the smallest summary of history needed to evaluate the consequences of the next action. A feature can improve prediction and still be irrelevant to the optimal action; another can have weak standalone predictive power yet matter because it changes risk or constraints.

1.3 Costs: benchmark, cash, and risk

Let P_t be the execution price for a child order. For a buy program, arrival- price implementation shortfall per share is

$$\text{IS} = \frac{1}{Q} \int_0^T v_t(P_t - m_0) dt + X_T \frac{P_T - m_0}{Q}.$$

The second term is an opportunity or terminal penalty if the order is not complete. In a model with mandatory completion it is zero, but removing it from the objective without enforcing $X_T = 0$ creates the absurd optimal policy of not trading.

Write the execution price schematically as

$$P_t = m_t + \frac{s_t}{2} + h(v_t, L_t),$$

for a buy, where h is instantaneous impact or liquidity cost and L_t denotes market conditions. The midprice may itself evolve as

$$dm_t = \mu_t dt + \sigma_t dW_t + g(v_t) dt + dJ_t.$$

The term $g(v_t)$ represents persistent impact and J_t jumps. This separation already exposes two distinct consequences of speed: trading faster can increase the paid liquidity cost h , while trading slower leaves inventory exposed to μ , σ , and signal decay.

A generic objective is

$$\min_{\pi} E^{\pi} \left[\int_0^T c(Z_t, u_t) dt + \Phi(Z_T) \right] + \lambda \mathcal{R}^{\pi},$$

where $\mathcal{R}$ is a risk functional. Common choices include variance, expected shortfall, exponential utility, a running inventory penalty $\int X_t^2 \sigma_t^2 dt$, or hard risk limits. These are not equivalent. Variance produces convenient linear-quadratic models; tail risk and hard constraints may produce qualitatively different policies.

1.4 Bellman’s principle

In discrete time, let the next state have transition law $p(z' | z, u)$. The value function is the minimum expected remaining cost:

$$V_{k(z)} = \min_{u \in \mathcal{U}(z)} \left\{ c_{k(z,u)} + E[V_{k+1}(Z_{k+1}) | Z_k = z, u_k = u] \right\}.$$

The terminal condition is $V_{N(z)} = \Phi(z)$. Bellman’s principle says that after any realized transition, the remaining actions must be optimal for the new state. The equation contains three inputs that empirical work must supply:

- a transition model for prices, liquidity, volume, fills, and impact state;
- a cost model for spread, fees, impact, risk, and non-completion; and
- a feasible action set representing urgency, participation, venue, and order- type constraints.

The continuous-time limit leads to a Hamilton–Jacobi–Bellman equation. For a diffusion $dZ_t = b(Z_t, u_t) dt + \Sigma(Z_t, u_t) dW_t$,

$$0 = \partial_t V + \min_u \left\{ c(z, u) + \nabla V^T b(z, u) + \frac{1}{2} \text{tr}(\Sigma \Sigma^T \nabla^2 V) \right\}.$$

The HJB is not the starting point; it is a compressed statement of the state, dynamics, objective, and information assumptions already chosen.

Control implication

A better forecast matters only through the Bellman comparison between actions. If all feasible actions have nearly the same conditional cost, large predictive gains can have little economic value. Conversely, a modestly predictive feature can be valuable near a switching boundary.

1.5 Prediction, causal response, and control

Three questions are often conflated:

Question	Target	Example
Prediction	$E[Y \mid X = x]$ under the observed policy	Expected slippage of orders similar to this one
Causal response	$E[Y(u) - Y(u')]$ under an intervention	Incremental cost of raising participation from 10% to 20%
Control	$\arg \min_{\pi} E[C^{\pi}]$	Adaptive schedule using impact, alpha, and liquidity state

Historical executions were selected by traders and algorithms. Urgent orders receive high participation; urgent orders may also arrive during adverse price moves. A regression of slippage on participation therefore mixes the response to speed with the reason speed was chosen. High out-of-sample predictive accuracy does not remove this endogeneity.

Research warning

Before interpreting a feature as an impact driver, draw the data-generating graph: latent urgency and alpha influence both the action and future prices; liquidity influences action, fills, and cost; early realized price moves can cause the policy to adapt. Conditioning on post-decision variables can create rather than remove bias.

For control, a purely predictive model can still be useful if deployed only within the support of the historical policy and validated on the decisions it changes. Structural or causal interpretation requires stronger design: randomization, valid instruments, natural experiments, explicit behavior-policy models, or credible assumptions about confounders.

1.6 A dimensional check

Dimensional analysis catches many modeling mistakes. If v is shares/minute, X shares, σ dollars/share/sqrt(minute), and the temporary-impact coefficient η satisfies $h(v) = \eta v$, then η has units dollars · minute/share². The cost rate $vh(v) = \eta v^2$ has units dollars/minute, and its integral has units dollars.

Normalizing quantity by ADV, time by a trading day, and cost by volatility can improve cross-sectional stability, but normalization does not make the data dimensionless by magic. State the reference horizon and whether volatility is daily, intraday, or annualized.

1.7 A research workflow derived from the control problem

1. Define decision time, benchmark, horizon, side, and unit of analysis.
2. Specify the action the model will change.
3. List state variables observable before that action.
4. State the counterfactual required: prediction under the current policy or response under an alternative action.
5. Build leakage-safe training and evaluation sets.
6. Estimate conditional cost and transition distributions with uncertainty.
7. solve the control problem and test sensitivity to estimation error.
8. Evaluate the complete policy, including fills, constraints, and opportunity cost—not just the model’s loss.

1.8 Interview checks

Interview check

Why is “minimize expected slippage” incomplete? State at least four missing choices. A good answer identifies the benchmark, completion constraint, information set, risk criterion, action space, and whether the expectation is predictive or interventional.

Interview check

A feature improves slippage R^2 but leaves the schedule unchanged. Is it valuable? Not necessarily. Explain this using action-value differences and discuss where the feature could still matter: constraints, tails, calibration, or regimes near a policy boundary.

Interview check

Why might realized participation be endogenous? Give one pre-trade and one within-order mechanism, then propose an empirical design that reduces each bias.

1.9 Exercises

1. Formulate a ten-bucket buy program with state (X_k, m_k, a_k, V_k) , action v_k , hard maximum participation, and a terminal non-completion penalty. Write the Bellman recursion.
2. Add a latent liquidity regime with noisy spread/depth observations. Define the belief-state update and explain why the last observation alone is not Markov.
3. Compare variance penalization with a 95% expected-shortfall objective. Give a scenario in which their optimal schedules differ.

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4. Draw a causal graph for slippage, realized participation, volatility, urgency, and short-horizon alpha. Identify one variable that is a confounder and one that may be post-treatment.
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2 Market Microstructure as the Controlled System

An execution model controls neither an abstract Brownian motion nor a daily return. It submits orders into a trading mechanism. The mechanism determines priority, fills, spread capture, information leakage, and how quickly liquidity recovers. Microstructure supplies the causal stories behind the state variables and cost functions of Chapter 1.

2.1 The limit-order book

At time t , let the best bid and ask be b_t and a_t . Define

$$m_t = \frac{a_t + b_t}{2}, \quad s_t = a_t - b_t.$$

A limit order supplies liquidity at a specified price; a marketable order demands liquidity and trades against resting orders. Price-time priority usually means better prices execute first and, at the same price, earlier orders execute first. Amendments may lose queue priority.

The displayed book is a schedule of marginal liquidity, not a promise that all depth remains available. A market order of size q walks the ask until its quantity is filled. If $A_{t(p)}$ is cumulative displayed ask depth through price p , the mechanical book cost relative to the mid is

$$C_{t(q)} = \int_0^q (P_t^{\text{ask}(x)} - m_t) dx,$$

where $P_t^{\text{ask}(x)}$ is the marginal ask price at cumulative quantity x . This is an **instantaneous** snapshot cost. It is not the same as the realized impact of an algorithm spread through time.

Intuition

Depth is a stock; volume is a flow. A book can look thin but replenish rapidly, or look deep and vanish when touched. Execution capacity depends on both the visible stock and the dynamics of replenishment.

2.2 Spread is compensation, state, and cost

The bid–ask spread compensates liquidity suppliers for several components:

- order-processing and venue costs;
- inventory risk while holding an unwanted position;
- adverse selection against better-informed or faster traders; and
- option value granted to a resting order while the market moves.

For an aggressive buy at the ask, the immediate mid-relative cost begins near $\frac{s_t}{2}$, before fees and impact. A passive buy may appear to earn $\frac{s_t}{2}$, but the fill is selected: it is more likely when sell pressure arrives or the fair price is moving down. Spread capture must therefore be evaluated after a suitable markout horizon and with non-fills included.

Definition: effective and realized spread

For a trade at price P_t with sign $\varepsilon_t \in \{-1, +1\}$, the effective spread is $2\varepsilon_t(P_t - m_t)$. A realized-spread measure replaces the contemporaneous mid with a future mid $m_{t+\tau}$: $2\varepsilon_t(P_t - m_{t+\tau})$. Their difference is a markout or adverse-selection component, conditional on the chosen horizon τ .

2.3 Order flow and price formation

Trade signs are persistent: large investors split orders, and correlated agents may act on related information. Yet returns are much less predictable than raw sign persistence suggests. Liquidity must adapt. The market may provide less depth, widen spreads, or adjust prices more strongly when order flow is surprising; predictable flow can have smaller marginal response.

A simple signed-flow model is

$$\Delta m_{t+1} = \beta \varepsilon_t f(q_t) + \gamma^T z_t + \xi_{t+1},$$

where z_t captures prior flow and liquidity. Interpreting β as “the impact of a trade” is dangerous unless trade initiation and market state are handled. The trade may convey information, remove a stale quote, mechanically deplete depth, or be caused by the same news that moves price.

The propagator representation makes path dependence explicit:

$$m_t = m_0 + \sum_{j < t} G(t-j) f(q_j) \varepsilon_j + M_t,$$

where $G(l)$ is a decay kernel and M_t is exogenous price innovation. A slowly decaying G means past flow remains part of the state. The model connects persistent flow, transient impact, and market resilience [1].

Control implication

If impact decays, remaining inventory is not a sufficient control state. The algorithm also needs a summary of its outstanding impact. Trading now changes both inventory and the cost of future trades.

2.4 Three meanings of “market impact”

The phrase is used for distinct objects:

1. **Mechanical response:** the price change caused by consuming or inducing a change in available liquidity, holding information fixed.
2. **Informational response:** prices move because the order reveals information or shares a cause with future returns.
3. **Reduced-form conditional cost:** the empirical slippage of executions with given characteristics under an observed policy.

All three matter economically. Only the first is a clean intervention effect, and even it depends on the intervention definition and horizon. A production cost model often estimates the third. It can be useful without supporting a structural claim, provided policy changes do not move too far outside the data.

2.5 Temporary, permanent, and transient response

A traditional execution model separates temporary impact, paid by the current child order, from permanent impact, which shifts all subsequent prices. Real markets more often show a response that builds during a metaorder and partially decays afterward. This motivates a transient state

$$dr_t = -\rho r_t dt + \kappa v_t dt, \quad P_t = m_t^0 + r_t + h(v_t),$$

where ρ is resilience and m_t^0 is an unaffected price. The solution

$$r_t = r_0 e^{-\rho t} + \kappa \int_0^t e^{-\rho(t-s)} v_s ds$$

shows why two schedules with the same total quantity can have different costs: their child orders overlap differently through the decay kernel.

The “unaffected price” is a modeling counterfactual, not an observable tape field. Estimating it is an identification problem.

2.6 Liquidity state and resilience

Useful liquidity measurements operate at several horizons:

Object	Examples	Control relevance
Price	spread, mid, microprice	aggression and benchmark cost
Book stock	depth, slope, imbalance	near-term capacity and fill risk
Order-flow dynamics	trades, adds, cancels, signed imbalance	adverse selection and depletion
Resilience	refill time, spread recovery, impact decay	spacing and pause decisions
Market activity	volume rate, trade count, event intensity	participation and clock speed

The microprice is one compressed top-of-book signal. If best-bid size is Q^b and best-ask size is Q^a , a common convention is

$$p^{\text{micro}} = \frac{aQ^b + bQ^a}{Q^a + Q^b}.$$

When bid depth dominates, the microprice lies closer to the ask. This is an empirical predictor, not a no-arbitrage fair value; queue cancellations, hidden liquidity, and deeper levels can reverse its meaning.

2.7 Clock choice

Calendar time is not the only clock. Trading intensity varies sharply through the day, so a minute near the close may contain far more events and volume than a minute at lunch.

- **Calendar time** is required for deadlines, alpha half-life, and latency.
- **Event time** advances with trades or book events and is useful for local microstructure dynamics.
- **Volume time** advances by market volume and often makes participation policies and intraday variance more homogeneous.

If cumulative expected volume is

$$\tau(t) = \int_0^t \frac{E[dV_s]}{E[V_T]},$$

then a uniform schedule in τ is a VWAP-style schedule in calendar time. Volume time helps remove seasonality but does not eliminate stochastic volume or selection: realized volume is not known at the start.

Research warning

Never use completed-day volume to normalize a feature used intraday unless it is explicitly a forecast or the task is retrospective measurement. Realized ADV-like denominators can leak future activity and make backtests too stable.

2.8 Intraday seasonality and the close

Spreads, depth, volatility, volume, and participant composition vary by time of day. Lower observed slippage near the close need not mean that time alone causes lower impact. Possible mechanisms include:

- greater volume and faster replenishment;
- different order mix and benchmark incentives;
- more predictable one-sided flows around the auction;
- selection of which orders remain or begin late;
- shorter markout horizons or benchmark construction; and
- conditioning on realized rather than forecast volume.

A time-of-day coefficient is a useful reduced-form adjustment, but research should test which state variable it proxies. Otherwise a regime change in the close can break the coefficient without warning.

2.9 Fragmentation, dark liquidity, and auctions

In a fragmented market, the action includes venue. Displayed prices may be similar while fill probability, queue length, fee/rebate, latency, odd-lot treatment, adverse selection, and hidden liquidity differ. A router faces a joint allocation and timing problem with censored feedback: for unchosen venues, the counterfactual fill is not observed.

Dark venues can reduce displayed information leakage and sometimes provide price improvement, but execution is uncertain and fills may be adversely selected. Closing auctions concentrate liquidity and eliminate continuous-book timing within the match, yet imbalance risk and benchmark incentives create a different control problem. An optimal policy must value optionality: waiting for uncertain passive or dark fills consumes time that could have been used to complete elsewhere.

2.10 From mechanism to model feature

A disciplined feature story has four links:

1. mechanism: why the variable should affect price, fill, or cost;
2. timing: when it becomes observable relative to the decision;
3. horizon: how long its information should persist;
4. action: which control choice should change.

For example, book imbalance may predict the next mid move over tens of events. It could justify a brief change in aggression, but it is unlikely to determine a two-hour schedule without an aggregation and persistence model.

2.11 Interview checks

Interview check

A passive fill earns half the spread. Why can its expected markout still be negative? Explain selection into fills, adverse flow, queue position, and why non-fills must be valued.

Interview check

Why might a thin displayed book have low realized impact? Discuss hidden liquidity, fast replenishment, cancellation behavior, child-order spacing, and the difference between snapshot depth and flow capacity.

Interview check

Observed slippage is lower near the close. What would you investigate before adding a time-of-day multiplier? Separate liquidity mechanisms, order selection, benchmark effects, denominator leakage, and post-trade horizon.

2.12 Exercises

1. Derive $p^{\text{micro}} - m$ in terms of spread and normalized top-level imbalance. Check its sign when bid depth exceeds ask depth.
2. Simulate the resilience state for two schedules with equal quantity: one block and ten equally spaced child orders. Compare the area under $v_t r_t$.
3. Propose a leakage-safe estimate of the intraday volume curve for a symbol with sparse history. Explain how hierarchical pooling or clustering would help.
4. Define a state and action space for a two-venue router with one lit and one dark venue. Identify the censored outcomes.

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3 Transaction Costs and Statistical Identification

Transaction-cost analysis is where market data becomes evidence. It is also where seemingly minor definitions can reverse conclusions. A benchmark chooses a counterfactual; a sampling rule chooses a population; a standard error chooses which dependence is acknowledged. Before fitting a sophisticated model, we must make these choices explicit.

3.1 The parent order is the unit of intent

Let a parent buy order have requested quantity Q , fills (q_i, p_i, t_i) , completed quantity $Q_f = \sum_i q_i$, and average execution price

$$\bar{P} = \frac{\sum_i q_i p_i}{Q_f}.$$

Child fills from the same parent are not independent research observations. They share urgency, benchmark, signal, market path, and adaptive policy. Treating each fill as independent inflates sample size and understates uncertainty.

Use a side variable $\varepsilon = +1$ for buys and -1 for sells. A side-adjusted cost relative to benchmark B is

$$C_B = 10^4 \varepsilon \frac{\bar{P} - B}{B} \text{ bps.}$$

Positive values then represent cost on both sides. State whether fees, rebates, commissions, taxes, and financing are included.

3.2 Benchmarks answer different questions

Benchmark	Question	Main vulnerability
Decision price	Cost from investment decision onward	Decision timestamp and pre-routing delay
Arrival price	Quality after algo receives the order	Pre-existing short-horizon alpha and arrival noise
Interval VWAP	Performance versus market volume schedule	Endogenous interval, realized-volume leakage, benchmark gaming
Close	Performance for close-valued portfolios	Large exogenous price move before the close
Reversion price	How much execution-period move persists	Arbitrary horizon and new information after completion

Arrival-price shortfall is natural for measuring the execution algorithm because it starts the clock when the algo gains control. It is not a pure impact measure: the no-trade price could have moved during the execution for informational or random reasons.

Intuition

A benchmark is not a neutral ruler. It specifies the world against which the execution is judged. Arrival asks, “What happened after control was handed to the algo?” VWAP asks, “Did the schedule beat the market’s realized volume- weighted path?” These are different objectives, not competing estimates of one true cost.

3.3 Implementation-shortfall decomposition

Suppose $Q_f \leq Q$ shares fill and mark unfilled inventory at price P_T . Total arrival shortfall in dollars for a buy is

$$\text{IS} = \sum_i q_{i(p_i - P_0)} + (Q - Q_f)(P_T - P_0) + \text{explicit costs}.$$

It can be decomposed algebraically in many ways. One useful research decomposition separates:

- spread and explicit-cost component;
- market movement during the schedule;
- price paid relative to contemporaneous mid;
- post-trade reversion or persistent markout; and
- opportunity cost of unfilled quantity.

The decomposition is descriptive unless an unaffected price path is identified. For example, contemporaneous-mid cost captures local aggression but the mid may already contain the program’s earlier impact.

3.4 The missing counterfactual

For order i , define potential cost $Y_{i(a)}$ under execution policy a . The causal effect of changing from policy a to a' is

$$\tau = E[Y_{i(a')} - Y_{i(a)}].$$

Only one potential outcome is observed. A historical comparison estimates $E[Y|A = a'] - E[Y|A = a]$, which equals τ only under assumptions such as exchangeability after conditioning on pre-treatment state X :

$$(Y(a), Y(a')) \perp A \mid X,$$

together with overlap and consistent treatment definition.

Execution challenges each assumption:

- urgency and alpha affect both chosen strategy and subsequent price;
- policy parameters vary continuously and adapt within the order;
- an algorithm can change which fills and even which completion times exist;
- one order’s action may affect another order’s outcome; and
- market regimes with one policy may have little overlap with another.

Research warning

Realized participation, duration, completion, and fill count are usually outcomes of the policy as well as descriptors of the order. Conditioning on them can block part of the effect or introduce collider bias. Prefer intended parameters and pre-trade forecasts when defining treatment and controls.

3.5 Randomized and quasi-experimental comparisons

Randomized A/B tests are the cleanest design when operationally safe. Randomize at the parent-order level, preserve allocation logs, and check:

1. treatment assignment occurred before any treatment-dependent action;
2. sample-ratio mismatch and logging failures;
3. pre-treatment balance by side, size, liquidity, urgency, symbol, and time;
4. adherence and crossovers;
5. interference between simultaneous orders; and
6. heterogeneous effects and tail risk, not merely mean cost.

If randomization is unavailable, options include matching, regression adjustment, inverse-propensity weighting, doubly robust estimation, instruments, and natural experiments. Each replaces randomization with assumptions. A method name does not make those assumptions credible.

For binary treatment with propensity $e(X) = P(A = 1|X)$, the inverse-weighted average-treatment estimator is

$$\hat{\tau} = \frac{1}{n} \sum_i \left[A_i \frac{Y_i}{e(X_i)} - (1 - A_i) \frac{Y_i}{1 - e(X_i)} \right].$$

Extreme weights reveal lack of overlap. Trimming changes the target population and must be reported.

3.6 Dependence and uncertainty

Order costs are heavy-tailed, heteroskedastic, and dependent across common symbols, days, clients, or market events. The standard error of an IID sample mean, $\frac{s}{\sqrt{n}}$, is too small if observations share shocks.

Choose a resampling or variance strategy aligned with the variation of interest:

- cluster by parent order when modeling child-level events;
- cluster or block by date for common market shocks;
- consider symbol and date multiway clustering for cross-sectional panels;
- use a block bootstrap for serially dependent time series; and
- report robust location and tail summaries alongside the mean.

If orders carry different notionals, distinguish the equal-order estimand from the dollar- or share-weighted estimand. A weighted mean answers a different business question and can be dominated by a few large orders.

3.7 A leakage-safe data contract

For every column, record:

Field	Timestamp	Question
feature time	when value became knowable	Could the live policy have used it?
decision time	when action was selected	Does the feature precede treatment?
event time	exchange or normalized market time	Are feeds synchronized?
label horizon	when outcome is complete	Do train/test labels overlap?
revision time	when corrected data arrived	Is research using a later revision?

Parent orders should be assigned wholly to train, validation, or test. Temporal splits better represent deployment than random row splits. Purge overlapping label windows when neighboring samples share future returns, and fit all normalizers, encoders, clusters, and feature selectors inside the training fold.

3.8 Modeling cost distributions

The conditional mean is often insufficient. A schedule may reduce average cost while worsening the right tail. Useful targets include

$$E[C|X], \quad \text{median}(C|X), \quad Q_{\tau(C|X)}, \quad P(C > c|X).$$

Quantile regression, distributional models, or conformal prediction can describe conditional uncertainty. Evaluation should include:

- calibration of intervals or quantiles;
- error by size, participation, spread, volatility, liquidity, side, and time;
- stability across symbols, dates, and regime shifts;
- economic loss under the downstream optimizer; and
- sensitivity to benchmark and markout horizon.

A low mean-squared error can coexist with systematically underestimated large- order cost—the exact region most important for control.

3.9 From TCA to model diagnosis

Residual analysis should ask structured economic questions. Let $e_i = C_i - \hat{C}_i$. Plot or tabulate conditional residual means and quantiles against:

- order size as a fraction of forecast ADV or depth;
- intended and realized participation;
- spread, volatility, volume, and order-flow imbalance;
- duration and time of day;
- sector, product, symbol liquidity, and futures roll proximity;
- client objective and benchmark; and
- market regime.

Patterns imply different failures. Residual curvature versus participation suggests functional-form error. Bias near rolls suggests a missing regime or contract-liquidity variable. Intraday structure after controlling for liquidity could indicate participant mix, benchmark incentives, or selection.

Control implication

TCA should close the control loop. The question is not merely whether average residual is zero, but whether errors are concentrated where the optimizer changes speed. Evaluate model error weighted by policy sensitivity.

3.10 Interview checks

Interview check

Algorithm B has 2 bps lower average arrival shortfall than A. What do you ask before declaring B better? Cover assignment, order mix, benchmark, completion, weighting, uncertainty, overlap, tails, and interference.

Interview check

Why should parent orders—not child fills—usually define the train/test split? Explain shared latent intent, market path, policy adaptation, and label leakage.

Interview check

Your model underestimates cost near futures rolls. What are plausible causes? Discuss migration between contracts, changing depth/volume denominators, spread and volatility, sample support, contract identity, and participant mix.

3.11 Exercises

1. Derive the relation between equal-order and notional-weighted average cost. Construct a two-order example in which they rank two algorithms oppositely.
2. Design an A/B test for two POV controllers. Specify the randomization unit, primary metric, guardrails, stratification variables, and interference check.
3. Build a causal graph containing urgency, alpha, intended participation, realized participation, market movement, and shortfall. Explain why realized duration can be a bad control.
4. Given repeated orders across 100 symbols and 40 dates, propose two uncertainty estimates and state what dependence each preserves.

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4 Market-Impact Models and Estimation

Market-impact estimation sits between measurement and control. The model must compress executions with different sizes, speeds, products, and states into a conditional cost surface. The optimizer will then differentiate or compare that surface outside the exact observations used to fit it. Shape, uncertainty, and identification matter as much as average predictive fit.

4.1 What is the estimand?

Before choosing a functional form, specify:

- response: arrival shortfall, VWAP slippage, contemporaneous-mid cost, or post-trade markout;
- horizon: execution interval and any reversion window;
- unit: dollars, basis points, ticks, spread units, or volatility units;
- population: parent orders, metaorders, child orders, or public trades;
- conditioning state: only information available at the intended prediction time; and
- interpretation: reduced-form prediction, policy response, or structural impact.

These choices define different targets. There is no benchmark-free scalar called “true impact.”

4.2 Scaling variables

Let

$$x = \frac{Q}{V}, \quad \rho = \frac{Q}{V_T T},$$

where Q is order size, V a daily-volume scale, and V_T the relevant market-volume rate. Then x is daily size fraction and ρ is participation. If volume rate is constant, duration satisfies

$$T = \frac{Q}{\rho V_T}.$$

Size, participation, and duration are therefore mechanically related. Including all three in a regression can create unstable interpretation; in real data the identity is noisy because volume varies and policies adapt, but the dependence does not disappear.

Costs are often normalized by volatility and liquidity. A stylized model is

$$E[C \mid X] = \sigma [\beta_0 + \beta_1 x^\delta \rho^\gamma] + \beta_s s + \beta^T z,$$

where $0 < \delta < 1$ represents concavity in size and z contains pre-trade state. A square-root specification uses $\delta = \frac{1}{2}$ [2]. This is an empirical scaling law over a population and range—not a universal theorem.

Intuition

Concavity says doubling size less than doubles per-share impact, not that total dollar cost is concave. If per-share cost scales as Q^δ , total cost scales as $Q^{1+\delta}$ and is convex for $\delta > 0$.

4.3 Shape and units

Suppose per-share temporary impact is $h(v) = \eta |v|^\gamma \text{sign}(v)$. For a constant-rate program, total temporary cost is

$$\int_0^T v_t h(v_t) dt = \eta Q^{1+\gamma} T^{-\gamma}.$$

Thus faster completion raises cost when $\gamma > 0$. If $0 < \gamma < 1$, per-share impact is concave in rate while total instantaneous cost $vh(v)$ remains convex. This distinction is important for optimization.

Empirical power laws are convenient because logs turn multiplicative structure into addition:

$$\log C = \alpha + \delta \log x + \gamma \log \rho + \theta^T z + \varepsilon.$$

But log regression changes the error model, excludes or transforms nonpositive costs, and estimates $E[\log C|X]$, not directly $\log E[C|X]$. Retransformation requires care under heteroskedasticity. Direct nonlinear least squares or a distributional likelihood may better match the deployment target.

4.4 Temporary, permanent, and decay models

4.4.1 Reduced-form schedule cost

A practical model predicts total cost from parent-order and market features:

$$C_i = f\left(\frac{Q_i}{\text{ADV}_i}, \rho_i, \sigma_i, s_i, t_i, z_i; \theta\right) + \varepsilon_i.$$

It is easy to serve and optimize, but it collapses the execution path. Changing the child-order pattern while holding summary features fixed may invalidate it.

4.4.2 Permanent-plus-temporary model

Let unaffected midprice follow $dS_t = \sigma dW_t$ and observed execution price be

$$P_t = S_t + \int_0^t g(v_s) ds + h(v_t).$$

Here g shifts later reference prices while h affects the current execution. This decomposition powers classical optimal execution, but observed post-trade prices do not uniquely reveal it without assumptions about the unaffected path.

4.4.3 Transient or propagator model

Let signed trading rate be v_t . A continuous-time propagator is

$$P_t = S_t + \int_0^t G(t-s) f(v_s) ds.$$

The expected impact cost is the quadratic-like path functional

$$C[v] = \int_0^T v_t \int_0^t G(t-s) f(v_s) ds dt.$$

This captures overlap and decay. Kernel and instantaneous-response shapes must obey restrictions to avoid price-manipulation strategies [3]. A model that fits observed paths but permits profitable round trips is unsuitable for control.

4.5 Data construction

A parent-order impact table should distinguish fields known at arrival from realized outcomes:

Stage	Examples	Use
Pre-trade	requested size, forecast ADV/ volume curve, spread, volatility, depth, alpha	features and stratification
Policy	intended urgency, POV bounds, benchmark, algo version	treatment and policy context
Within-order	realized participation, fills, pauses, market path	diagnostics or sequential model
Post-trade	shortfall, markouts, completion, reversion	labels and decomposition

Use pre-trade denominators for a deployable pre-trade estimator. If a post-trade TCA model uses realized volume, label it as retrospective and do not silently pass the same formula to a live optimizer.

Filter rules also define the estimand. Excluding high-volatility days, incomplete orders, rolls, or difficult symbols can make residuals look better by removing the decisions for which the model is most needed. Prefer explicit regime flags, censoring models, and separate support diagnostics.

4.6 Constrained nonlinear estimation

Let y_i be measured cost and $f(x_i; \theta)$ a nonlinear impact curve. Estimate

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \sum_i w_i \rho_{\text{loss}}(y_i - f(x_i; \theta)),$$

where Θ enforces economically motivated bounds such as positive scale and $0 < \delta < 1$. The loss may be squared, Huber, or a likelihood-based deviance. Trust-region reflective algorithms are useful for bound-constrained nonlinear least squares, but an optimizer's convergence flag is not evidence that the model is identified.

Check:

1. multiple starting points and boundary solutions;
2. parameter-profile or bootstrap uncertainty;
3. Jacobian conditioning and correlated parameters;
4. stability across time, symbols, size ranges, and sides;
5. monotonicity and curvature over the deployment domain; and
6. extrapolated marginal costs used by the controller.

If residual variance grows with order size, unweighted least squares prioritizes large absolute errors. That may be economically appropriate—or it may cause the model to ignore typical orders. Choose weights to match the target estimand, then report performance both weighted and unweighted.

Research warning

Shape constraints can encode sensible extrapolation, but they can also conceal misspecification. If many estimates hit a concavity bound, investigate missing interactions, regimes, denominator error, and action endogeneity before interpreting the bound as an empirical discovery.

4.7 Endogeneity of execution speed

Consider

$$C_i = \beta \rho_i^\gamma + \theta^T X_i + \varepsilon_i.$$

Urgency U_i may raise ρ_i and correlate with future adverse movement in ε_i . Liquidity shocks during the order can raise both realized participation difficulty and cost. Adaptive algorithms may speed up after price moves, reversing the apparent causal direction. Therefore $E[\varepsilon_i | \rho_i, X_i] \neq 0$ is plausible.

Strategies include:

- randomized parameter perturbations within safe bounds;
- intent-to-treat analysis using assigned rather than realized speed;
- instruments that shift speed without directly shifting cost;
- fixed effects or rich pre-trade controls when conditional exchangeability is credible;
- sequential causal methods for time-varying treatment and confounding; and
- structural models jointly describing policy and outcomes.

None is automatic. A valid instrument is an economic argument, not just a variable correlated with participation.

4.8 Hierarchical pooling

Symbol-specific models are noisy for illiquid products; a universal model can miss persistent heterogeneity. A hierarchical specification partially pools:

$$\beta_j \sim \mathcal{N}(\mu_\beta, \tau_\beta^2), \quad C_{i,j} = \beta_j f(X_{i,j}; \theta) + \varepsilon_{i,j}.$$

Liquid symbols with abundant data learn their own coefficients; sparse symbols shrink toward the cross-sectional mean or a cluster mean. Features describing tick size, spread, sector, market cap, and liquidity can explain part of the hierarchy.

Clustering volume profiles or impact residual shapes can help, but clusters must be fitted on training history and monitored for membership instability. A soft hierarchy often expresses uncertainty better than assigning a sparse symbol to one hard cluster.

4.9 Machine learning: where it helps

Flexible ML models can capture interactions among spread, volatility, imbalance, time, and liquidity. Useful patterns include:

- a structural baseline plus an ML residual correction;
- generalized additive models for interpretable nonlinear effects;
- boosted trees for tabular interactions;
- quantile or distributional models for tail-aware control;
- multitask or hierarchical models across products; and
- sequence/state-space models for within-order adaptation.

A hybrid model

$$\hat{C}(x) = C_{\text{struct}(x; \hat{\theta})} + f_{\text{ML}(x)}$$

preserves scaling intuition while allowing empirical corrections. Constrain or audit the combined surface for monotonicity, smooth marginal costs, support, and policy stability. A jagged prediction surface can create unstable bang-bang schedules even if its average test error is low.

Control implication

Model selection should include a **policy loss**: solve the downstream control problem using each candidate, then evaluate the resulting actions under a common simulator, randomized experiment, or credible off-policy method. The best predictor is not always the best decision model.

4.10 Validation as a surface, not a score

Report at least:

- mean error, MAE/RMSE, and tail or quantile calibration;
- residual maps over size $\times$ participation and volatility $\times$ spread;
- chronological and regime holdouts;
- symbol/client/product-group stability;
- uncertainty of both predictions and parameters;
- monotonicity/concavity violations and extrapolation distance; and
- induced schedule changes and sensitivity.

A useful local sensitivity is

$$S_{j(x)} = \partial \hat{C} \frac{x}{\partial} x_j.$$

Compare S_j with empirical local effects and with the optimizer's response. If small coefficient changes produce large policy changes, the control problem needs regularization, uncertainty-aware optimization, or tighter constraints.

4.11 Interview checks

Interview check

Why can a concave per-share impact curve still produce a convex execution problem? Multiply per-share impact by traded quantity or rate and show the resulting exponent.

Interview check

Your test RMSE improves, but optimal schedules become erratic. Why? Discuss derivative noise, extrapolation, weak overlap, unconstrained interactions, and policy-sensitive rather than prediction-average validation.

Interview check

How would you test whether lower close-period slippage is a time effect or a liquidity-composition effect? Propose pre-trade controls, within-symbol/date comparisons, interactions, overlap checks, and a policy or natural-experiment design.

4.12 Exercises

1. Derive total temporary cost for $h(v) = \eta v^\gamma$ under constant rate and determine when it is convex in Q and in v .

-
2. Fit mentally a power law with δ near its upper bound. List four reasons besides “impact is linear” that could generate the boundary estimate.
 3. Design a hierarchical model for futures contracts around rolls. Specify which parameters pool by product, front/next contract, and days-to-roll.
 4. Compare a structural-plus-residual model with an unconstrained boosted tree. Define a validation suite focused on the schedule each model induces.
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5 Optimal Liquidation and Participation Control

We now close the loop: use an impact model and a price-risk model to choose a schedule. Classical optimal execution is valuable not because its assumptions are literally complete, but because it isolates the central trade-off and makes comparative statics transparent. It is the reference model against which more realistic controllers should be understood.

5.1 Why an optimum exists

For a buy order, trading very fast reduces exposure to adverse price movement and alpha decay but consumes liquidity aggressively. Trading very slowly lowers instantaneous impact but leaves inventory at risk and may miss the deadline. An interior optimum arises when marginal urgency benefit equals marginal liquidity cost.

Intuition

Execution speed is an insurance decision. Faster trading pays a more certain liquidity premium to reduce an uncertain future-price and non-completion exposure. Risk aversion, adverse alpha, and short deadlines buy more insurance.

5.2 A discrete control model

Divide the horizon into N intervals. Let x_k be inventory before interval k , $n_k = x_k - x_{k+1}$ shares traded, and $x_0 = Q$, $x_N = 0$. Suppose

$$S_{k+1} = S_k + \sigma\sqrt{\Delta t}\xi_{k+1} + \gamma n_k,$$

and execution price is

$$P_k = S_k + h\left(\frac{n_k}{\Delta}t\right).$$

The first equation contains persistent impact and random price movement; the second adds temporary liquidity cost. With quadratic temporary cost and a variance penalty, the problem becomes a convex quadratic program. Position, participation, trading-rate, and completion constraints can be added directly.

Bertsimas and Lo formulate execution as a dynamic optimization of expected cost [4]. Almgren and Chriss make the expected-cost/risk frontier explicit [5]. Their exact conventions differ, but both demonstrate that optimal execution is a control problem, not simply a forecast of cost.

5.3 Continuous Almgren–Chriss derivation

Let remaining inventory X_t satisfy $\dot{X}_t = -v_t$, with $X_0 = Q$ and $X_T = 0$. Assume an unaffected arithmetic Brownian price

$$dS_t = \sigma dW_t$$

and linear temporary impact $h(v) = \eta v$. Ignoring path-independent terms, the mean–variance objective is

$$J[X] = \int_0^T [\eta \cdot X_t^2 + \lambda \sigma^2 X_t^2] dt.$$

The first term is expected temporary cost because $v = -\cdot X$; the second is the running variance exposure of unexecuted inventory. The Euler–Lagrange equation is

$$\eta \ddot{X}_t = \lambda \sigma^2 X_t.$$

Define $\kappa = \sqrt{\lambda \frac{\sigma^2}{\eta}}$. Applying the boundary conditions gives

$$X_t = Q \frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)},$$

and

$$v_t = Q \kappa \frac{\cosh(\kappa(T-t))}{\sinh(\kappa T)}.$$

The schedule is front-loaded. As $\lambda \rightarrow 0$, $\kappa \rightarrow 0$ and $X_t \rightarrow Q(1 - \frac{t}{T})$: the risk-neutral optimum is constant rate. As risk aversion or volatility increases, or temporary impact falls, κ increases and the program trades earlier.

The characteristic execution time is approximately $\frac{1}{\kappa}$. This single ratio exposes the economics:

$$\text{urgency} \sim \kappa = \sqrt{\frac{\text{risk pressure}}{\text{liquidity-cost curvature}}}.$$

5.4 What permanent impact does

With linear permanent impact, the price shift is proportional to cumulative quantity traded. For fixed total quantity, its expected cost is proportional to $\frac{Q^2}{2}$ and is path-independent. It changes expected total cost but not the optimal schedule in the basic model.

This surprising result is a good diagnostic. If a proposed “permanent-impact” term changes the fixed-quantity schedule, check whether it is actually nonlinear, state dependent, asymmetric, or mixed with a temporary component. Nonlinear permanent impact can also admit manipulation and requires no-arbitrage analysis.

5.5 Efficient frontier

Every admissible schedule has expected cost E and variance V . Varying λ traces an efficient frontier: lower variance generally requires higher expected impact cost. A desk’s urgency setting is therefore an implicit exchange rate between expected basis points and execution uncertainty.

Mean–variance is convenient but has limitations. Cost distributions are not Gaussian; impact parameters are uncertain; jumps and fill risk affect tails; and clients may care about benchmark underperformance probabilities rather than variance. Expected shortfall, utility, or chance constraints can replace the quadratic penalty, usually at the cost of closed forms.

5.6 Adding alpha or adverse drift

Suppose the expected price drift relevant to a buy is α_t dollars per share per unit time. Integration by parts shows that expected market-movement cost contributes

$$\int_0^T \alpha_t X_t dt.$$

The objective becomes

$$J[X] = \int_0^T [\eta \cdot X_t^2 + \lambda \sigma^2 X_t^2 + \alpha_t X_t] dt.$$

Positive α_t means waiting is expected to be costly and induces earlier trading; negative drift rewards patience. The Euler equation is now forced:

$$\eta \ddot{X}_t = \lambda \sigma^2 X_t + \frac{\alpha_t}{2}.$$

If alpha decays, for example $\alpha_t = \alpha_0 e^{-\omega t}$, only its value over the remaining horizon matters. A fast-decaying signal favors capturing value early; an uncertain signal should be shrunk and incorporated with estimation risk rather than treated as known.

Research warning

Alpha supplied in “volatility units” needs a horizon. A dimensionless score is not a drift until it is mapped to an expected price path and calibrated out of sample. Overconfident alpha estimates can dominate impact and create extreme schedules.

5.7 Nonlinear temporary impact

Let per-share temporary impact be $h(v) = \eta v^\gamma$ for $v \geq 0$. The running dollar cost is $\eta v^{1+\gamma}$. With risk,

$$J[X] = \int_0^T [\eta (-\cdot X_t)^{1+\gamma} + \lambda \sigma^2 X_t^2] dt.$$

The first-order condition is nonlinear. Closed form may disappear, but convexity is preserved for $\gamma > 0$. Numerical dynamic programming, direct collocation, or convex optimization can solve the discretized problem. Concave per-share impact does not imply a nonconvex total-cost objective.

When the estimated exponent is weak or uncertain, the optimal rate can be highly sensitive because the controller uses marginal rather than average cost:

$$\frac{d}{dv}[vh(v)] = \eta(1 + \gamma)v^\gamma.$$

This is why impact-surface derivatives deserve direct validation.

5.8 Volume time and POV control

Let forecast cumulative market volume be $M(t)$, with volume rate $\cdot M(t)$. A participation policy trades

$$v_t = \rho_t \cdot M(t).$$

Inventory follows

$$X_t = Q - \int_0^t \rho_s \cdot M(s) ds.$$

A constant ρ is uniform in forecast volume time. If $\cdot M(t)$ has a U-shape, the calendar-time schedule naturally trades more near the open and close.

An optimal POV controller chooses ρ_t by balancing:

- the impact curve’s marginal cost of participation;

- expected alpha loss from leaving inventory;
- price and benchmark risk;
- forecast liquidity and its uncertainty;
- minimum/maximum participation and rate limits; and
- terminal completion or residual penalty.

A generic discrete formulation over buckets is

$$\min_{q_1, \dots, q_N} \sum_k [C_{k(q_k; \hat{V}_k, Z_k)} + \alpha_k X_k \Delta t + \lambda \sigma_k^2 X_k^2 \Delta t] + \Phi(X_{N+1})$$

subject to

$$X_{k+1} = X_k - q_k, \quad 0 \leq q_k \leq \rho_{\max} \hat{V}_k, \quad X_1 = Q.$$

Here $\hat{V}_k$ must be a forecast at the decision time. Using realized bucket volume in an ex ante optimizer leaks information.

Control implication

A POV is not intrinsically “fast” or “slow.” Its calendar-time speed is the product of participation and market volume. The same 10% POV can be gentle in an active market and unable to complete in a quiet one.

5.9 Time-of-day liquidity and the apparent close advantage

Let temporary-cost scale vary by bucket, η_k . If estimated η_k is lower near the close because volume is high and spread is narrow, a cost-only optimizer back-loads. But several forces oppose it:

- inventory is exposed longer to price and benchmark risk;
- alpha may decay;
- close liquidity is forecast with error;
- auction access and continuous trading differ;
- maximum participation can make late completion infeasible; and
- the historical close coefficient may reflect selected order mix.

The right response to counterintuitive lower close slippage is not to reject it or hard-code it. Decompose the mechanism, use pre-trade state, test stability, and let terminal feasibility and risk compete with the estimated liquidity benefit.

5.10 Receding-horizon control

A static schedule uses forecasts available at arrival. A model-predictive controller repeats:

1. observe current inventory, fills, prices, volume, and liquidity;
2. update forecasts and impact/resilience state;
3. solve the remaining-horizon problem;
4. execute only the next action; and
5. roll forward.

This handles forecast error and changing conditions while retaining transparent optimization. It also creates a statistical complication: realized actions are endogenous to earlier shocks. Evaluation should compare complete policies, not regress cost naively on their realized controls.

Safeguards include rate-of-change limits, minimum time between major updates, uncertainty buffers, fallback schedules, and terminal feasibility checks. These are part of the policy, not engineering details outside the model.

5.11 Sensitivity and model risk

For the classical solution,

$$\kappa = \sqrt{\lambda} \frac{\sigma}{\sqrt{\eta}}.$$

Thus a 1% local increase in volatility or risk-aversion square root raises urgency approximately proportionally, while a 1% increase in η lowers κ by about 0.5%. This parameter sensitivity does not directly equal schedule sensitivity when κT is very small or very large.

Stress the policy under:

- impact coefficient and exponent uncertainty;
- volume forecast error by bucket and whole day;
- volatility and spread shocks;
- alpha sign error and faster/slower decay;
- roll, auction, and low-liquidity regimes;
- partial fills, outages, and delayed market data; and
- objective mismatch between arrival, VWAP, and close.

A robust controller may optimize worst-case or distributionally perturbed cost. A simpler and often effective approach is shrinkage, conservative liquidity forecasts, constraints, and receding-horizon updates.

5.12 Interview checks

Interview check

Why is the risk-neutral Almgren–Chriss schedule TWAP-like? Show that with quadratic rate cost and no drift or state variation, Jensen’s inequality or the Euler equation favors a constant rate.

Interview check

What happens when volatility doubles? In the basic model κ doubles, but describe the actual schedule change using κT and distinguish this from doubling total cost.

Interview check

A 25M order trades in a stock with 80M ADV. Should it front-load? There is no answer from size alone. Discuss horizon, participation capacity, spread, volatility, liquidity curve, alpha sign/decay, benchmark, impact curvature, and completion risk.

Interview check

Why can a pause reduce future cost in a transient-impact model but not in the basic temporary-impact model? Identify the added resilience state and its decay during the pause; then state the opportunity cost of waiting.

5.13 Exercises

1. Re-derive the hyperbolic-sine inventory path from the Euler–Lagrange equation and recover the linear schedule as $\lambda \rightarrow 0$.
 2. Prove with Jensen’s inequality that a constant rate minimizes $\int_0^T v_t^{1+\gamma} dt$ for fixed $\int v_t dt = Q$ and $\gamma > 0$.
 3. Add bucket-specific η_k and volume caps to a five-period quadratic model. Write its matrix quadratic-program form.
 4. Create three alpha scenarios—momentum, mean reversion, and zero—and sketch how each changes the inventory path.
 5. Design an MPC update for an order that falls behind its volume schedule at midday. List state updates, forecast changes, constraints, and fallback conditions.
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